



How Does Google Maps Know There Is Traffic Ahead?

A JOURNEY BEHIND THE TECHNOLOGY  | FROM DATA TO DESTINATION



Satellites



Millions
of Phones



Live & Historical
Traffic Data



AI &
Prediction



Better
Routes for You

SAME ROADS. SMARTER JOURNEYS.

*"Maps the World"
Data Tells the Story*



Heavy traffic ahead
18 min delay

Presented by
Daksh Khandelwal
Founder, ThinkHub

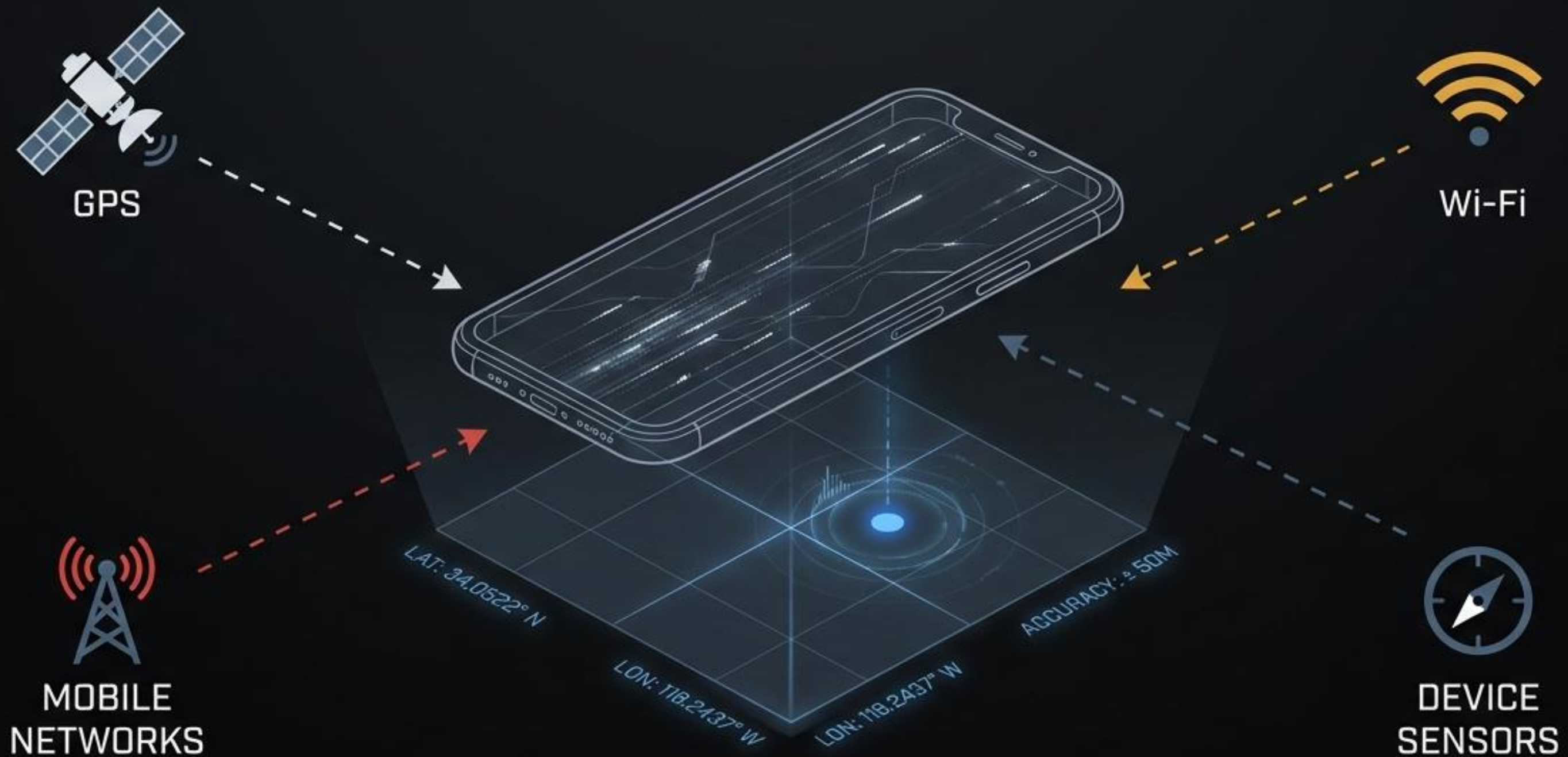
*Let's
Explore*

HOW DOES MAPS KNOW?

Heavy traffic ahead — **18 min delay**



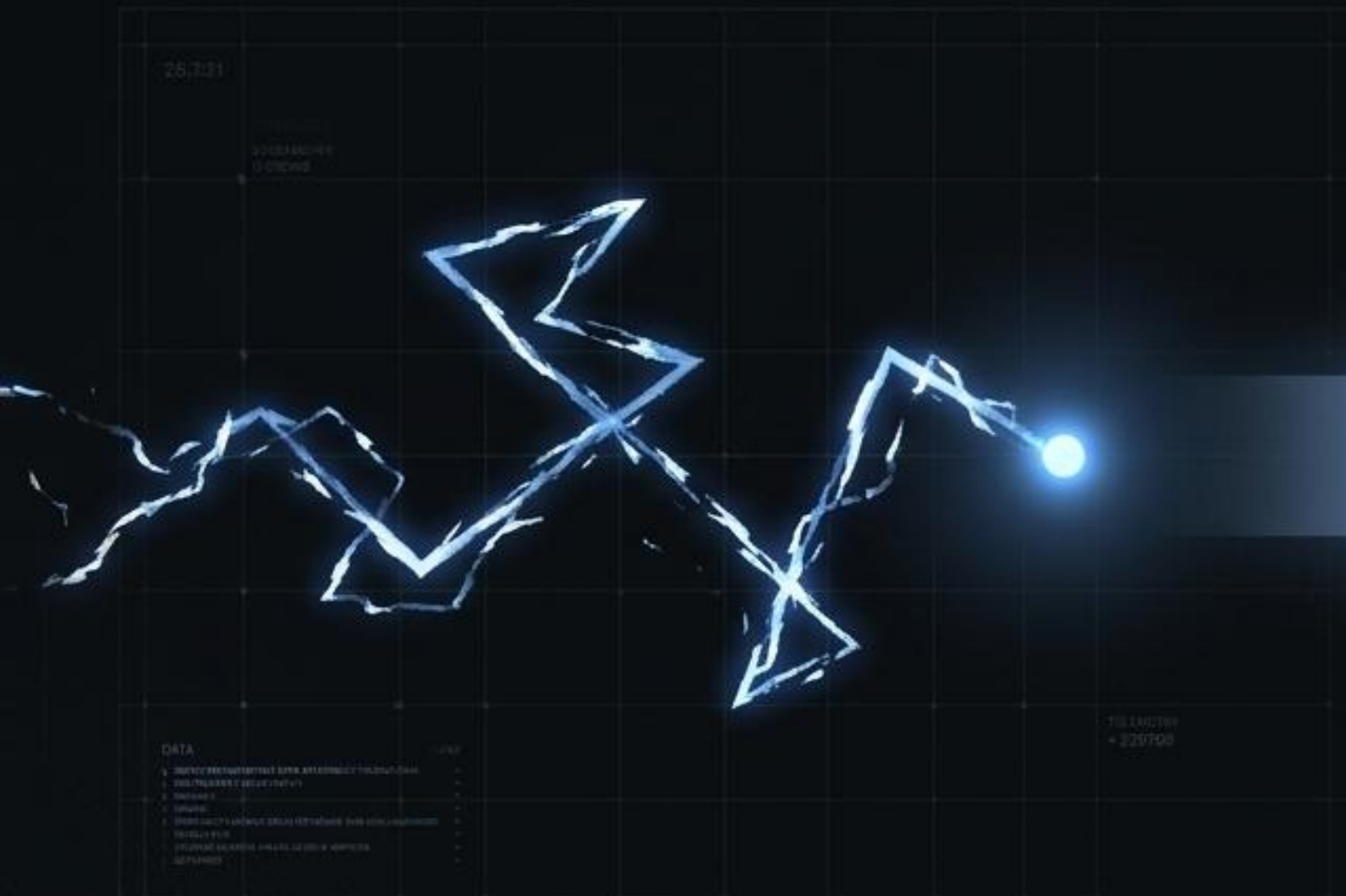
LOCATION IS AN ESTIMATE



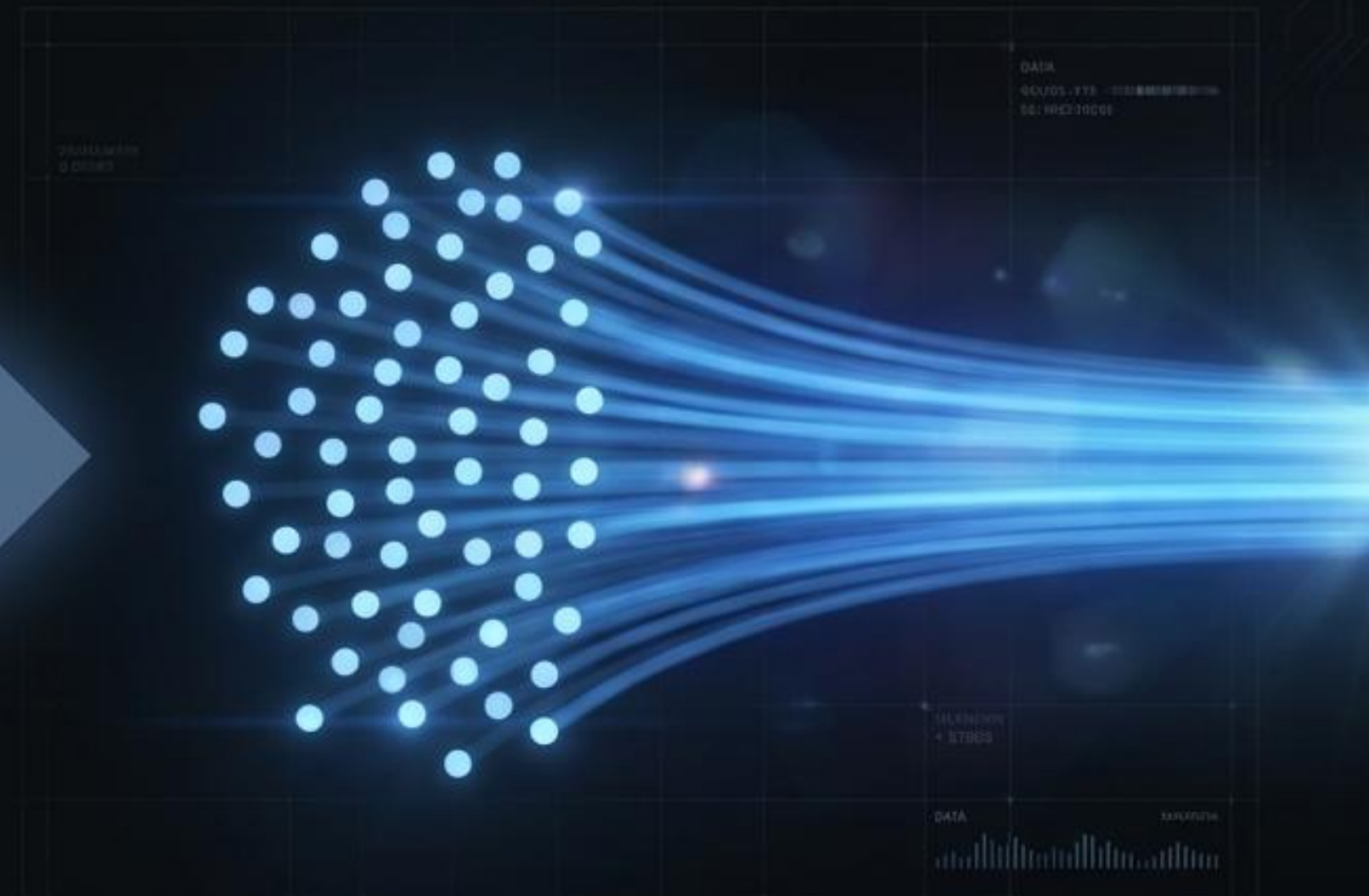
WHERE + WHEN = MOVEMENT



ONE DEVICE IS NOISE. MANY DEVICES FORM A PATTERN.



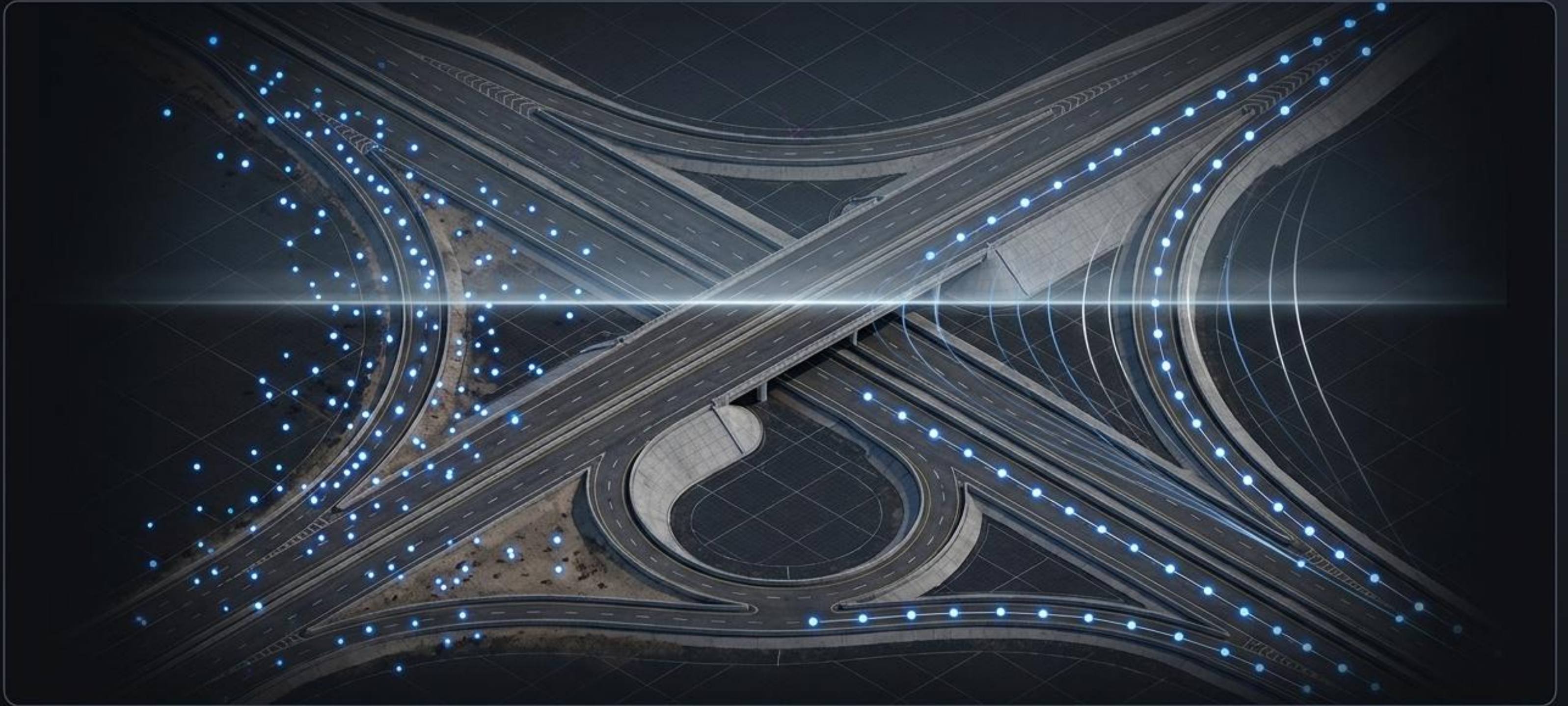
⚠ UNRELIABLE NOISE



➡ RELIABLE TRAFFIC PATTERN

DOTS BECOME ROADS

MAP MATCHING



Annotated Flow

1. Raw Coordinate: Latitude X,
Longitude Y.

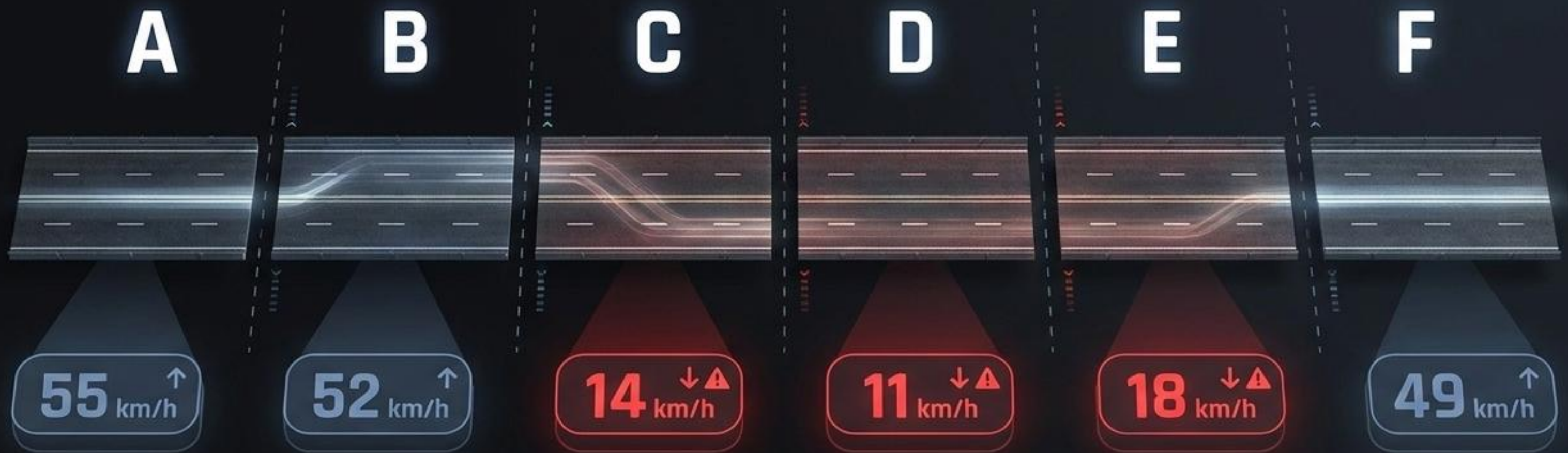


2. Context Engine: Is the car
on the highway, the parallel
service road, or the flyover?



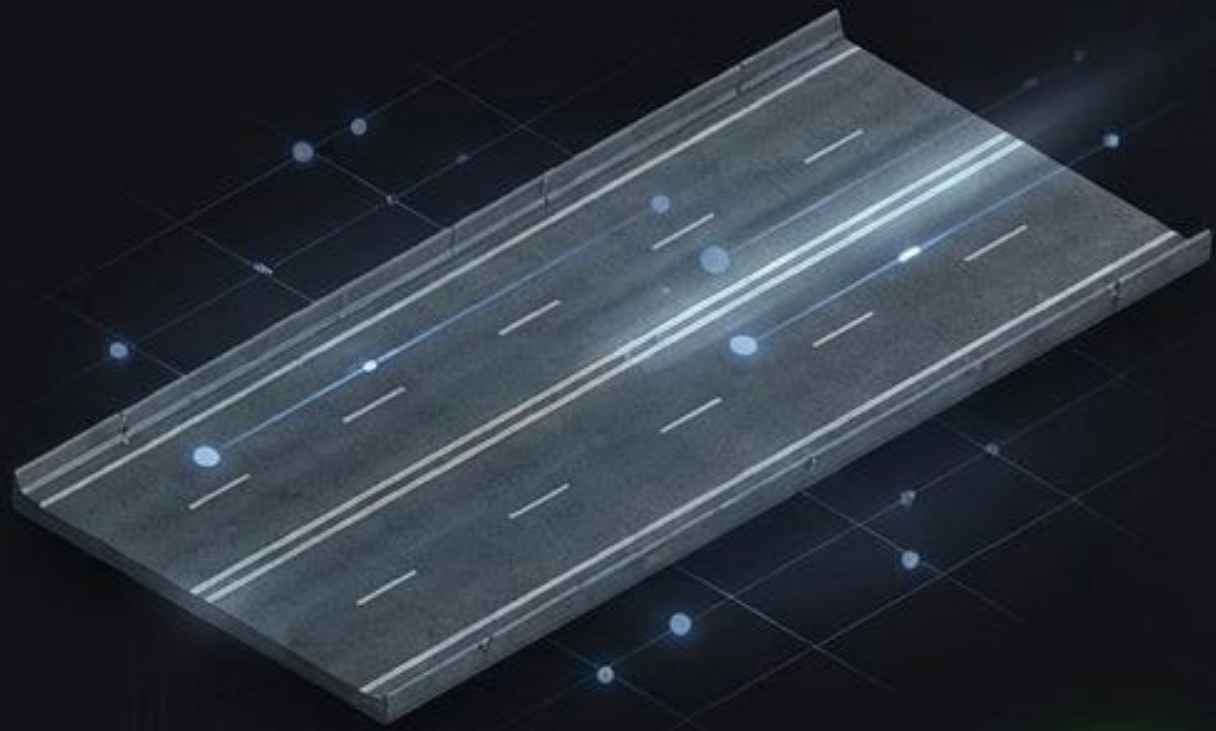
3. The Match: Algorithms snap
the location observations to
the exact underlying road
segment based on trajectory
and speed.

THE ROAD BECOMES DATA



DETECTING ANOMALIES

NORMAL TRAFFIC FLOW

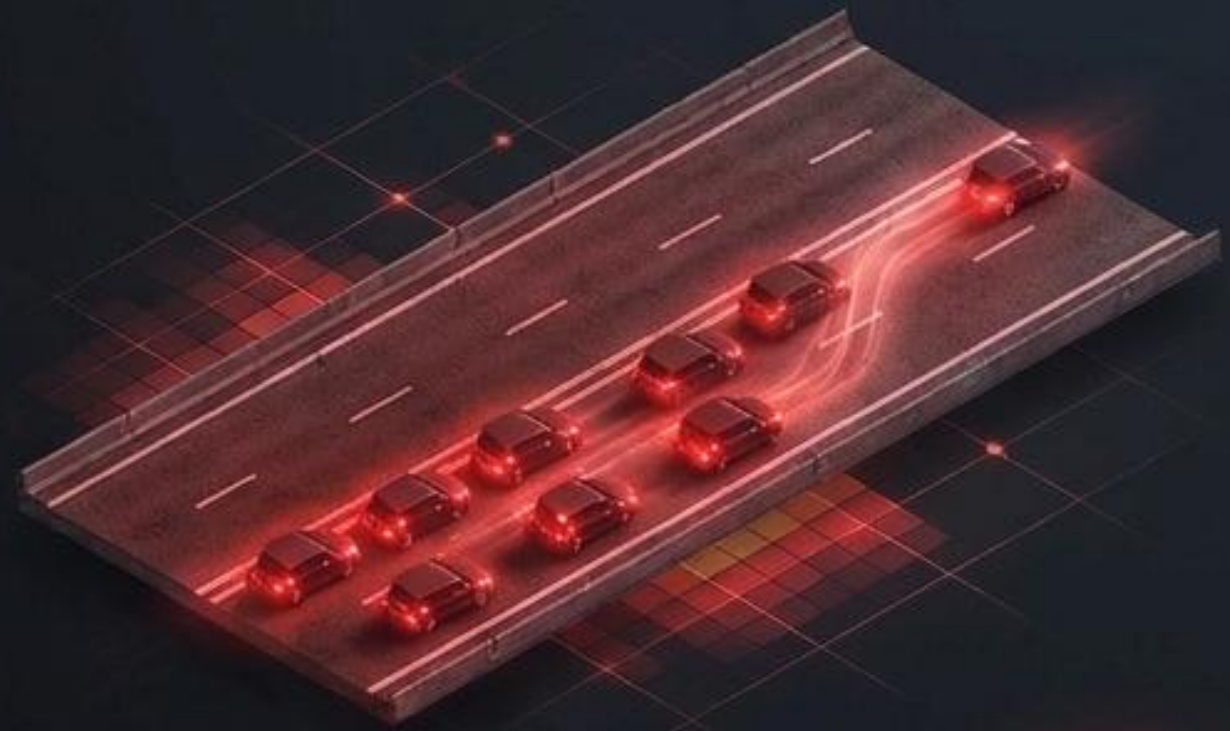


Telemetry

Expected: 50 km/h | Current: 48 km/h



CONGESTION ANOMALY



Telemetry

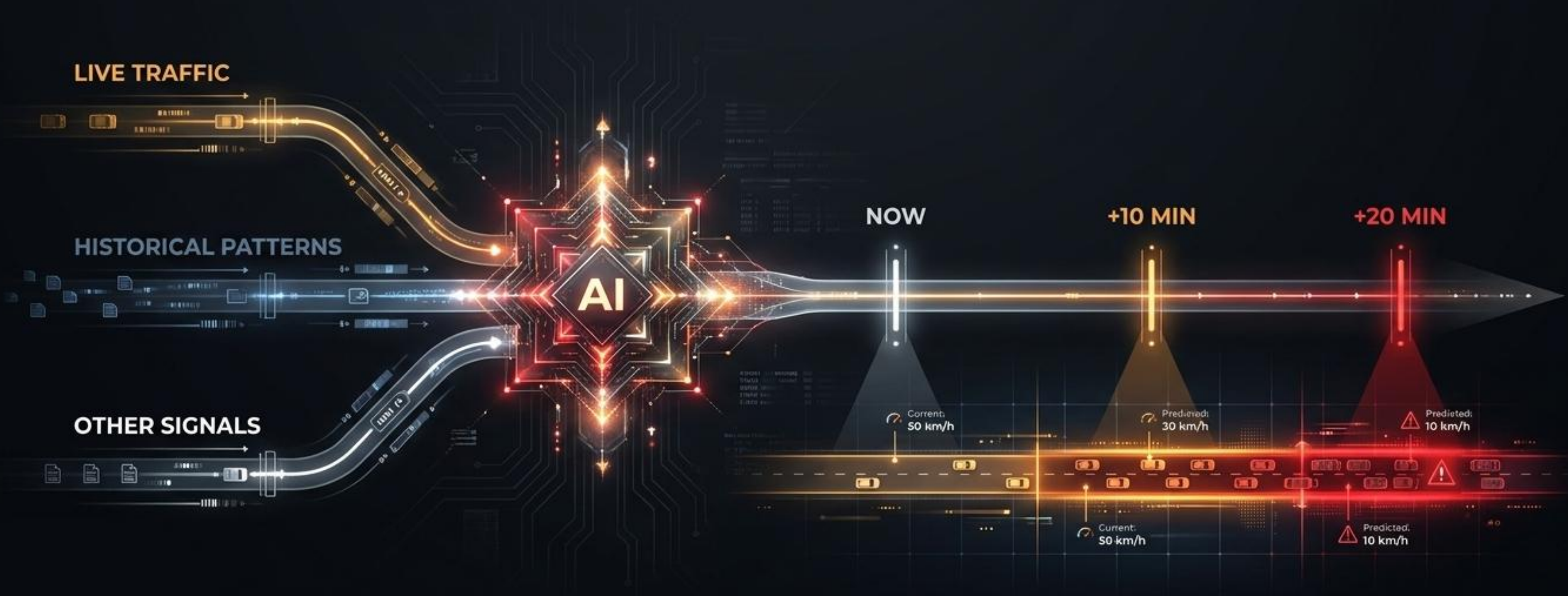
Expected: 50 km/h | Current: 12 km/h



THE PAST EXPLAINS THE PRESENT



DETECTION ≠ PREDICTION



Detection

Question

What is happening right now?

Data Source

Live Aggregated Movement.

Prediction

Question

What will happen in 20 minutes when you get there?

Data Source

Live Data + Historical
Baselines + Machine Learning.

Routing

Question

Given the prediction, what is the optimal path?

Data Source

Optimization Algorithms.

THE NETWORK IS A GIANT GRAPH

Nodes = Intersections |
Edges = Road Segments

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Edges = Road Segments

SLOW TRAFFIC DETECTED



HISTORICAL CONTEXT



Rush Hour

USER REPORTS



Community Disruption Alerts

AUTHORITATIVE DATA



Official Road Closure Inputs

DATA TELLS WHAT. SIGNALS TELL WHY.

ETA IS A PREDICTED JOURNEY



THE MAP KEEPS LEARNING



PREDICTION IS NOT CERTAINTY



IT DOESN'T SEE THE FUTURE. IT PREDICTS IT.





THANK YOU!

The future belongs to those who **build it continuously.**



START TODAY.
Take the first step.



STAY CONSISTENT.
Small steps, big results.



BUILD ONE PROJECT
at a time.



PRESENTED BY
DAKSH KHANDELWAL
FOUNDER, THINKHUB



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KEEP LEARNING. KEEP BUILDING. KEEP GROWING.